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3. If m x a + n x b is an invariant for the assignment a, b : = a + 8, b + 7, the values of m and n are

   (a) m = 8, n = 7 (b) m = 7, n = -8 (c) m = 7, n = 8 (d) m = 8, n = -7

4. Which of the following is not an invariant of the assignment?

   m, n := m+2, n+3

   (a) m mod 2 (b) n mod 3 (c) 3 X m - 2 X n (d) 2 X m - 3 X n

5. If Fibonacci number is defined recursively as

$$\mathbf{F(n)=}\begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ F(n-1)+F(n-2) & \text{otherwise} \end{cases}$$

   to evaluate F(4), how many times F() is applied?

   (a) 3 (b) 4 (c) 8 (d) 9

6. Using this recursive definition

$$\mathbf{a^{n}=}\begin{cases} 1 & \text{if } n = 0 \\ a \times a^{n-1} & \text{otherwise} \end{cases}$$

   how many multiplications are needed to calculate a^{10}?

   (a) 11 (b) 10 (c) 9 d) 8

Part II

1. What is an invariant?
2. Define a loop invariant.
3. Does testing the loop condition affect the loop invariant? Why?
4. What is the relationship between loop invariant, loop condition and the input-output recursively
5. What is recursive problem solving?
6. Define factorial of a natural number recursively.